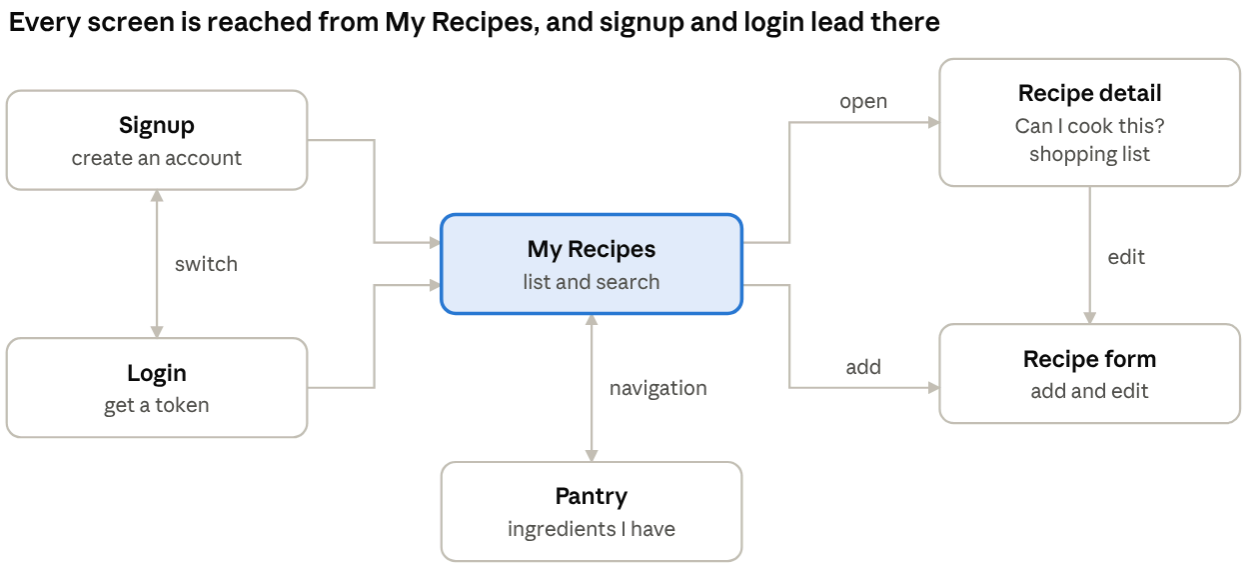
Tabkha – Milestone 1: Wireframes and Database Schema

Nov 8, 2026 · @clara

Tabkha has six screens and three collections, and every screen's data lives in User, Recipe or PantryItem.

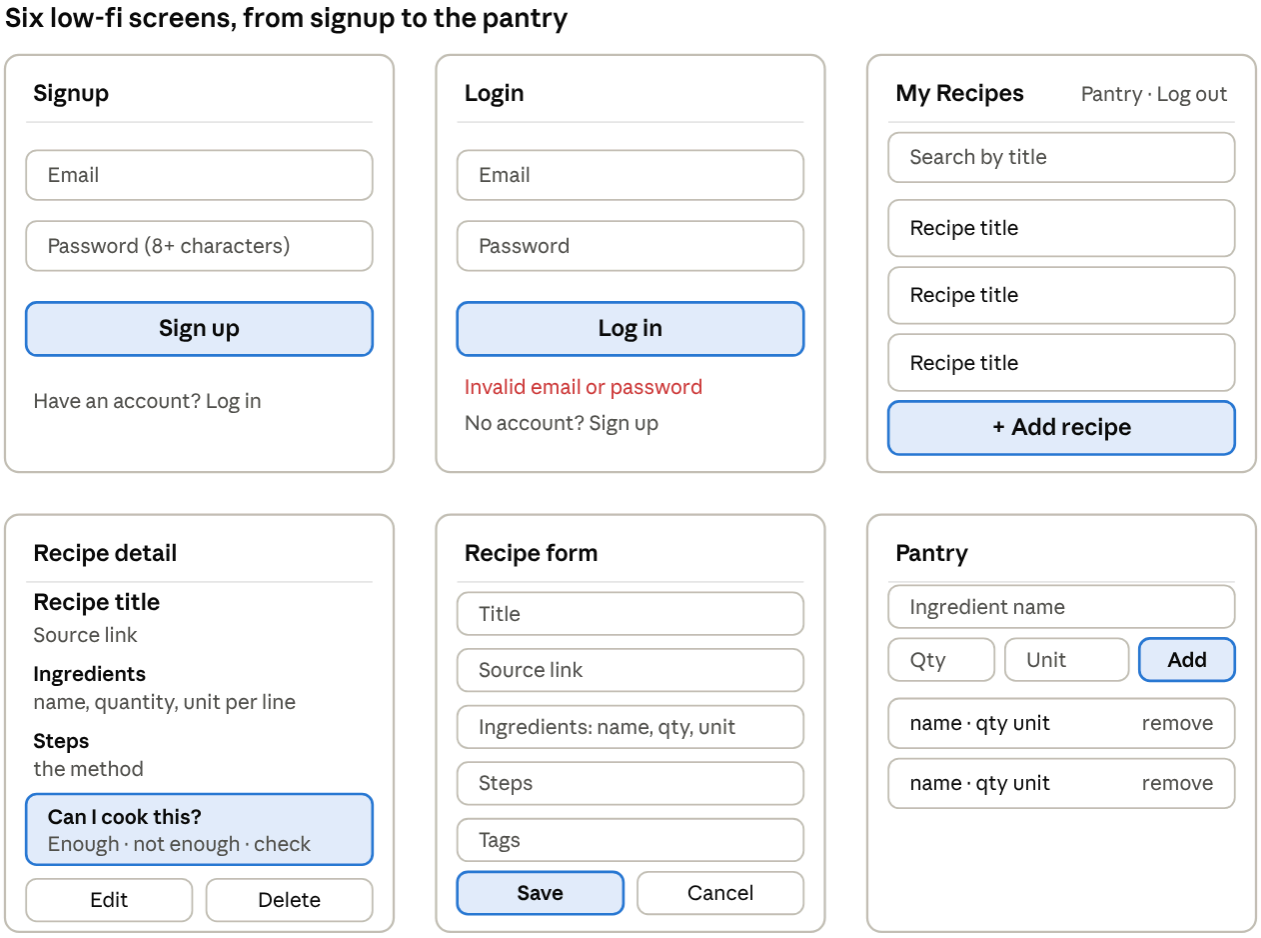
Screen flow



screen flow · 6 screens

Signup and login both lead to My Recipes, which opens a recipe, the recipe form and the pantry. The arrows show how a user moves between screens.

Wireframes



wireframes · 6 screens

Low-fi sketches of the six screens. The colored box marks the main action on each screen, and the red line under Login is the error message from the Session 19 login route.

Database schema

Three collections: User, Recipe and PantryItem. Recipe and PantryItem each point to their owner with `userId`, marked **ref** below.

Collection	Field	Type	Notes
User	`email`	String	required, unique, lowercase, trim; used to log in
	`passwordHash`	String	required; the bcrypt hash, never the plain password

Collection	Field	Type	Notes
	`createdAt`, `updatedAt`	Date	added automatically by `timestamps`
Recipe	`userId`	ObjectId, **ref** User	required; the owner, taken from the login token
	`title`	String	required, trim, max length 120
	`sourceUrl`	String	default empty; link to the original post
	`ingredients`	[{name, quantity, unit}]	default empty list; each item has a name, a quantity and a unit (three rows below)
	ingredients.name	String	required, trim; compared with the pantry name
	ingredients.quantity	Number	required, greater than 0
	ingredients.unit	String	required; one of g, kg, ml, l, piece, tbsp, tsp (checked by the server)
	`steps`	String	default empty; the method
	`tags`	[String]	default empty list
	`createdAt`, `updatedAt`	Date	added automatically by `timestamps`
PantryItem	`userId`	ObjectId, **ref** User	required; the owner
	`name`	String	required, trim; compared with recipe ingredients
	quantity	Number	required, greater than 0; what the user has
	unit	String	required; same fixed list as the recipe units
	`createdAt`, `updatedAt`	Date	added automatically by `timestamps`

Ingredients and tags are embedded in the recipe, because they are small and always read with it. PantryItem is its own collection, because it is used on its own and its list can grow (the Session 18 rule: embed or reference). The shape follows the Session 18 Note model, plus the Session 19 `userId` reference.

"Can I cook this?" is worked out when needed and never stored. For each recipe ingredient, find the pantry item with the same name (trimmed, lowercase). Same unit and enough: enough. Same unit and less: not enough, and the difference goes on the shopping list. Different unit: check the quantity yourself. No pantry item: missing. The recipe can be cooked only when every ingredient is enough.

Where each screen's data lives

The Milestone 1 check: every screen's data has a home, and no screen needs a fourth collection.

Screen	Data it shows or saves	Collection
Signup	email and password in, a new account out	User
Login	email and password checked, a token returned	User
My Recipes	the logged-in user's recipes, searchable by title	Recipe, filtered by `userId`
Recipe detail	one recipe, plus "Can I cook this?" and the shopping list of missing and short ingredients	Recipe and PantryItem; the shopping list is worked out when needed, not stored
Recipe form (add and edit)	title, source link, ingredient rows (name, quantity, unit), steps, tags	Recipe
Pantry	the ingredients the user has at home, with quantity and unit: list, add, remove	PantryItem, filtered by `userId`